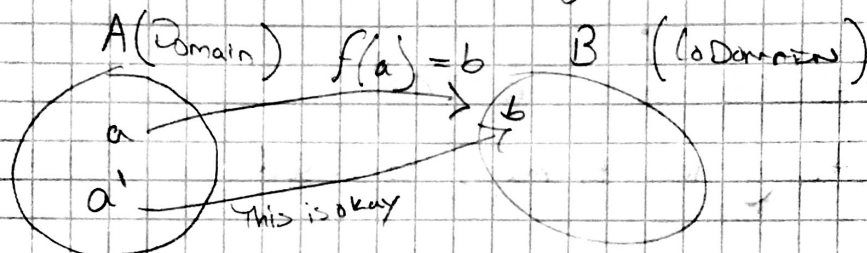


Functions

MAY 31

Functions
Sequences
Matrices
Cardinality
Algorithms

Defn: A function from a set A to B , written $f: A \rightarrow B$ is a rule that associates with each element of A a unique element.



A function should be defined for all elements in its domain

Defn: A set of ordered pairs, no two of which have the same 1st element

Defn b is the image of a

Defn a is the preimage of b

eg) Suppose $C \subseteq A$, then $f(C) = \{t : \exists c \in C \text{ with } f(c) = t\}$

eg) Suppose $f(x) = x^2$, then $f(\{1, 2, 3\}) = \{1, 4, 9\}$

Defn: The range is any $b \in B$ s.t. $f(a) = b$. The set of images

Injective / One-to-One: if $a \neq b$, then $f(a) \neq f(b)$

Contra for proving: $f(a) = f(b)$, then $a = b$

Surjective / Onto:

The range is equivalent to the codomain

$$\forall b \in B \exists a \in A \text{ s.t. } f(a) = b$$

Bijection is both one-to-one AND Onto

eg) consider $f: \mathbb{Z} \rightarrow \mathbb{Z}$ AND $g: \mathbb{N} \rightarrow \mathbb{N}$

$$f(x) = x + 1$$

$$g(x) = x + 1$$

one to one

ONLY ONE TO ONE

AND
ONTO

Inverse functions

Defn if $f: A \rightarrow B$, AND f is 1 to 1 AND onto
then $f^{-1}: B \rightarrow A$ is a function where $f^{-1}(b) = a$ and $f(a) = b$

Defn Identity functions $i: A \rightarrow A$

$\hookrightarrow$ everything in A maps to itself.

Operations on functions

$$(f+g)(x) = f(x) + g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

...
Same for rest of arithmetic operators.

Defn Let $g: A \rightarrow B$ and $f: B \rightarrow C$
Then the composition of $g \circ f: A \rightarrow C$

$$(g \circ f)(A) = g(f(A)) = C$$

Defn If f^{-1} exists, the $f \circ f^{-1} = f^{-1} \circ f$

(eg) Suppose $f(x) = 2x + 3$

Soln

$$y = 2x + 3$$

$$\frac{y-3}{2} = x = f^{-1}(y)$$

$$\text{Then } f^{-1}(y) = x$$

(eg) $(f \circ g)(x) = (g^{-1} \circ f^{-1})(y)$

Sequences + Summations

3

DefN: A function whose domain $\mathbb{N}$ (Natural Positive Numbers)

Notation: $f(n) \equiv a_n$

$$a_n = n^2 \quad n = 1, 2, 3, \dots$$

DefN Geometric Progression: has a common ratio, you achieve every $n+1$ term with a common ratio.

$$a, \text{ or, } ar^2, ar^3, ar^4 \dots 1, 2, 4, 16 \dots$$

DefN Arithmetic Progression: "Common difference"

$$a, a+d, a+2d \dots 3, 7, 11, 15 \dots$$

DefN Recursive Progression: The next term depends on earlier terms

eg) Suppose $\sum_{j=2}^7 a_j$ then $a_2 + a_3 + a_4 + a_5 + a_6 + a_7$

• The Sigma represents "Sum" from $j \rightarrow n$

eg) $\sum_{i=1}^5 i! = 1! + 2! + 3! + 4! + 5! = 153$

eg) $\sum_{i=1}^5 i = 1 + 2 + 3 + 4 + 5$

eg) $\sum_{i=1}^5 i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

eg) $\sum_{i=1}^5 7 = 7 + 7 + 7 + 7 + 7 = 35$

eg) $\sum_{i=1}^n ax_i + by_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n y_i$

→ In finite sums the common rules of arithmetic apply

(eg)

$$1 + r + r^2 + r^3 \dots r^n = \frac{r^{n+1} - 1}{r - 1}$$

• prove with Algebra

(4)

(eg)

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Matrices

• Used for large sequences of equations

Defn = A rectangular array of something rows and columns

• we can add, subtract, multiply, "divide" $m \times n$
 must be same size

must have $m \times k, k, n$ structure

$$(m \times k) \cdot (k \times n) \rightarrow (m \times n)$$

(eg)

$$\begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & 4 \end{pmatrix}^{(2 \times 3)} \cdot \begin{pmatrix} 1 & 4 & 2 \\ 2 & 1 & 5 \\ 3 & 4 & 1 \end{pmatrix}^{(3 \times 3)} = \begin{pmatrix} 1(1) + (3 \cdot 2) + (2 \cdot 3) & 4 + 3 + 8 & 4 + 10 + 2 \\ 0 + 2 + 12 & 0 + 1 + 16 & 0 + 5 + 4 \end{pmatrix}^{(2 \times 3)}$$

$$= \begin{pmatrix} 13 & 15 & 16 \\ 14 & 17 & 9 \end{pmatrix}$$

WARNING

if you have square matrices, $A \times B$ might
 not equal $B \times A$

Defn

Identity Matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

or

• if a matrix has an inverse, then $A \times A^{-1} = I_{\text{identity}}$

eg

Claim $\mathbb{Z} = \{ \dots -1, 0, 1, 2, \dots \}$ is countable.

- 1 $\rightarrow$ 0
- 2 $\rightarrow$ 1
- 3 $\rightarrow$ -1
- 4 $\rightarrow$ 2
- ...

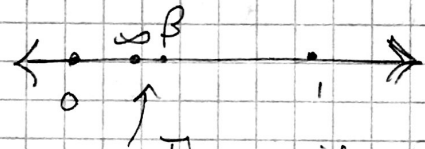
$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \% 2 = 0 \\ -(\frac{n-1}{2}) & \text{if } n \% 2 = 1 \end{cases}$$

eg

$\text{card}(\mathbb{N}) = \text{card}(\mathbb{Z})$ from the above example

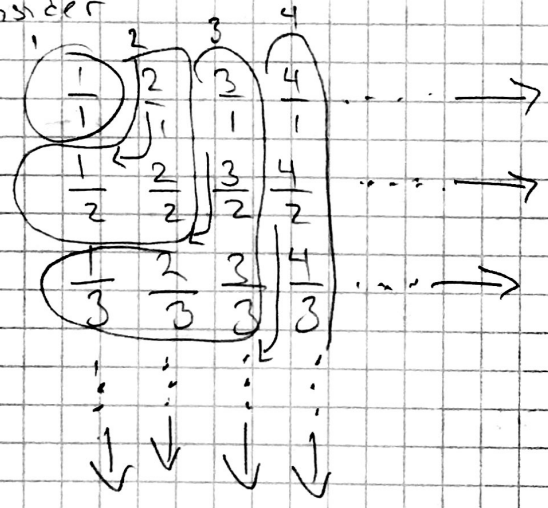
eg

Center: The set of rational numbers is countable



There exists a rational (infinite ratio) in fact between any α and β

Consider



eg

Center $\mathbb{R}$ is uncountable, in fact $(0, 1)$ is uncountable
 $\hookrightarrow$ Proof by Contradiction

- Suppose $(0, 1)$ is countable. then you can list all the real numbers $r_1, r_2, r_3, r_4, \dots$
- Then suppose $r_1 = .012345\dots$ $r_2 = .01010101\dots$
 $r_3 = .3219782\dots$

CARDINALITY

• Size of a set, or the number of elements in a set.

5

eg $|\{1, 2, 3, 4\}| = 4$

eg When do two sets have the same cardinality?

• When two sets have one-to-one correspondence, if there is a bijection between them.

eg What about infinite sets?

Def N A set is infinite if it is not finite.

Def N A set is finite if:

• it is empty set

• $\exists n \in \mathbb{N}$ s.t. S can be put in 1 to 1 correspondence with $\{1, 2, 3, \dots, n\}$

eg if $\text{card}(A) \leq \text{card}(B) \wedge \text{card}(B) \leq \text{card}(A)$
then $\text{card}(A) = \text{card}(B)$

eg $\mathbb{N} = \{1, 2, 3, \dots\}$ $2\mathbb{N} = \{2, 4, 6, 8, \dots\}$

$$f: \mathbb{N} \rightarrow 2\mathbb{N}$$

$$f(n) = 2n$$

Therefore, there are as many even integers as integers

Def N Countable: A set is countable if it is finite or if it can be into 1-1 correspondence with $\mathbb{N}$.

Algorithms

Defn

finite set of instructions that halts

8

Rating algorithms; we use the rate of growth of the function

Let $f, g: \mathbb{N} \rightarrow \mathbb{R}^+$, let those be functions and f is $O(g)$ ("f is big-o of g")

if $\exists$ constants c, k s.t. $f(x) \leq c g(x) \forall x \geq k$

- We're saying that g is eventually bigger than f .
- Sometimes c, k are called witnesses

ex

$f(x) = x^2 + 2x + 1$ is $O(x^2)$

Choose $c=4$ then at $k=1$, then every time $k \geq 1$, it is larger than $f(x)$

$$x^2 + x + x + 1 \leq 4x^2$$

ex

is $n^2 O(n)$? $\rightarrow$ No, no matter the c or k ,

you will never have a linear equation that dominates a quadratic

- Cantor proposed that he has a number that is not on the list you've created (7)

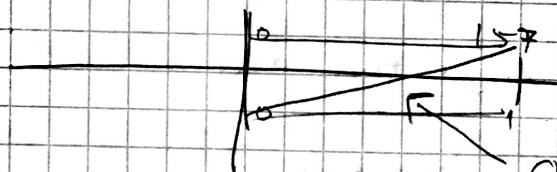
$$r_c = d_1 d_2 d_3 d_4 \dots \quad d_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ of } i \text{ is not } 1 \\ 2 & \text{otherwise} \end{cases}$$

- Therefore, any i^{th} position number in your list, the r_c will differ in the i^{th} place of the r_c number.

eg

Any two closed intervals considering $\mathbb{R}$ have the same cardinality.

$$|(0, 1)| = |(0, 157)|$$



one to one correspondence

Def N

$$|\mathbb{N}| = \aleph_0 \leftarrow \text{aleph null}$$

$$|\mathbb{R}| = \mathfrak{c} \leftarrow \text{continuum}$$

$$|\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} = \mathfrak{c}$$

• There is 1-1 between $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$

- because you can map the $\mathcal{P}$ set with binary, using bitwise functions

f is $O(g)$ if $\exists$ constants $c+k$ such that

$$f(x) \leq c g(x) \quad \forall x \geq k \quad (g \text{ eventually dominates } f)$$

(eg) $x^2 + 2x + 1$ is $O(n^2)$

x^2 is not $O(n)$

Facts

• All n^{th} degree polynomials are $\Theta(x^n)$

(eg) $a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$ $\leftarrow$ we really only care about the a_nx^n term.

• Exponential functions dominate all powers

(eg) no matter the x^n , any b^n will dominate any x^n

• because logarithmic bases are changed with constants, we don't really care about whether a logarithmic function is base 2, 10, 5 etc...

(eg) $O(\log n)$ will always be dominated by exponentials and polynomials

"Arithmetic of Complexity"

if $f_1 = O(g_1)$

what is? $f_1 + f_2$

$f_2 = O(g_2)$

$f_1 \cdot f_2$

Sum = $O(\max(g_1, g_2))$

Product = $O(g_1 g_2)$

$\hookrightarrow$ actually multiply them together.

• Big Omega and Big Theta

(2)

Ω

if $f = \Omega(g) \rightarrow f$ dominates g

if $f = \Theta(g) \rightarrow f$ dominates g & g dominates f

Finding a max in a list $\equiv O(n)$

Linear search $\equiv O(n)$

Matrix multiplication $\equiv O(n^3)$

Binary search $\equiv O(\log n)$

Bubble Sort $\equiv O(n^2)$
 Quick Sort $\equiv O(n \log n)$
 Traveling Salesman $\equiv O(n!)$
 1st Table Building $\equiv O(2^n)$
 Determining Parity $\equiv O(1)$
 (even or odd)

Integers

Defn if $d \neq 0$ $d | n$ "d divides n" or "d is a divisor of n"

if and only if there exists an integer k $n = d \cdot k$

$\Leftrightarrow \exists k \in \mathbb{Z} \ni n = d \cdot k$

Facts if $A | B \rightarrow a \in b$ "if a divides b" $\rightarrow \dots$

if $A | B \wedge A | C \rightarrow A | B + C$

if $A | B \wedge B | C \rightarrow A | C$

Division Algorithm Let a be an integer, let d be a positive integer

then $\exists ! q, r$ such that $n = dq + r$ with $0 \leq r < d$

Defn

$a \% b$ is the remainder you get when you divide a by b

(8)

$17 \bmod 3$ is 2

Defn

$$a \equiv b \pmod{n} \text{ or } [a \equiv_n b]$$

if a & b are ~~are~~ have the same mod value

(9)

$$17 \equiv_3 29$$

(ec)

$$17 \equiv_3 31$$

$$\rightarrow m \mid a - b$$

$$17 \% 3 \equiv 29 \% 3$$

(ec)

what is $(5 + 17 + 29) \bmod 3$? $\equiv \emptyset$

$$(5 \% 3 + 17 \% 3 + 29 \% 3) \% 3 \equiv \emptyset$$

$$2 + 2 + 2 \equiv_3 \emptyset$$

(ec)

Prove that in any collection of any 3 consecutive integers there is a multiple of 3.

If

let n be the smallest integer in this collection of 3 integers

By the division algorithm, n has to look like $3k$, $3k-1$, or $3k+2$

if $n = 3k$, then n is a multiple of 3

if $n = 3k+1$, then $n+2 = 3k+3$

② Prove every odd square is congruent to 1 mod 8. ④

meaning if $a \equiv 1 \pmod{8}$

Means they have the same remainder

• if n is odd then $n^2 = 8k+1$ ← want to show this

if n is odd then $(2j+1)^2 = 8k+1$

$$4j^2 + 4j + 1 = 8k + 1$$

Now we need to show that $4j^2 + 4j$ is a multiple of 8

$$4(j(j+1))$$

A pair of consecutive integers, thus it will evaluate to some factor of 8.

Representation of Integers → Positional Number System

Binary base 2 (0-1)

1 bit = 2 numbers

Octal base 8 (0-7)

3 bits = 4 numbers

Hexadecimal base 16 (0-15)

4 bits = 8 numbers

③

Decimal → Octal

$375_{10} \rightarrow 567$

← work

10 bits = 2^{10} numbers

divide, then use remainders

$$\begin{array}{r} 46 \\ 8 \overline{) 375} \\ \underline{8 \cdot 46} \\ 7 \end{array} \quad r=7$$

$$\begin{array}{r} 5 \\ 8 \overline{) 146} \\ \underline{8 \cdot 18} \\ 6 \end{array} \quad r=6$$

$$\begin{array}{r} 0 \\ 8 \overline{) 5} \\ \underline{8 \cdot 0} \\ 5 \end{array} \quad r=5$$

OCTAL → Decimal

$$372_8 = 192 + 56 + 2 = 150_{10}$$

$$64 \cdot 3 + 7 \cdot 8 + 2 \cdot 1$$

$$375 = 8 \cdot 46 + 7$$

$$375 = 8[8 \cdot 5 + 6] + 7$$

$$375 = 8(8 \cdot (6 \cdot 5 + 5) + 6) + 7$$

(5)

Binary Addition $O(n)$
 Binary multiplication $O(n^2)$

$$\begin{array}{r}
 11011 \\
 \times 1101 \\
 \hline
 11011 \\
 11011 \\
 11011 \\
 11011 \\
 \hline
 101001111
 \end{array}$$

Exponentiation

$$5^{23} \quad O(\log n)$$

(2)

$$\begin{array}{r}
 5^2 \quad \checkmark \\
 5^4 \quad \checkmark \\
 5^8 \quad \times \\
 5^{16} \quad \checkmark
 \end{array}$$

(1)

Because 23 is a 5 bit number, we want 5 squarings

(3)

$$5 \cdot 5^2 \cdot 5^4 \cdot 5^{16} = 5^{23}$$

Primes

Defn

A positive integer greater than 1 p is prime IFF the only divisors of p , are 1 and p .

FT Arithmetic

: every positive integer > 1 can uniquely factor into a product of prime

(eg)

is 201 prime? if the sum of the digits is a multiple of 3, its not prime

soln: NO

is 323 prime? Only have to test up to $\sqrt{323}$

soln: NO

FACT

There are infinitely many primes (countable)

Proof Euclid's proof by contradiction

- Assume that primes are finite in many.

$$- \text{Then } p_1, p_2, p_3, \dots, p_k$$

- $\prod_{i=1}^k p_i + 1$ if This number isn't a prime, then how is it possible that it has no prime divisors?

$$\text{--- Twin primes } \stackrel{eg}{=} 29, 31$$

• primes with 1 number between.

• The gap is 2 or less

Prime Number Theorem

There is a function

$$\pi(x) = \text{number of primes} \leq x$$

$$\pi(20) = 8$$

what about $\pi(1,000,000) \approx \frac{1,000,000}{\ln(1,000,000)}$ gives an approximate value

• As n gets bigger, this method becomes more accurate.

Euclidean Algorithm

(7)

Defn

GCD, greatest common divisor.

(a, b)

largest integer d s.t. $d|a \wedge d|b$

eg

$$\text{GCD}(300, 90) = \frac{\text{Solve}}{300} = 2^2 \cdot 3 \cdot 5^2 \quad 90 = 2 \cdot 3^2 \cdot 5$$

$$\text{Then } 2 \cdot 3 \cdot 5 = 30$$

Defn

LCM Smallest integer that is a multiple of a, b

eg

$$\text{LCM}(300, 90) = 2^2 \cdot 3^2 \cdot 5^2 = 900$$

eg

$$\text{GCD}(287, 91)$$

we can use Euclidean Algorithm

$$\begin{array}{r} \textcircled{1} \quad 91 \overline{) 287} \quad \textcircled{2} \quad 14 \overline{) 91} \quad \textcircled{3} \quad 7 \overline{) 14} \\ \underline{14} \quad \quad \quad \underline{14} \quad \quad \quad \underline{14} \\ 14 \quad \quad \quad 7 \quad \quad \quad 0 \end{array} \quad \leftarrow \text{STOP AT } R=0$$

Look at the last remainder, then that's the GCD.

$$\equiv 7$$

If $\text{GCD}(a, b) \equiv 1$, then they are relatively prime

↳ if you execute Euclid's Algorithm and get 1, then they are relatively prime

(8)

Linear Congruences

(Congruent)

↳ Same remainder
when mod in

$$3x \equiv 14 \pmod{10}$$

$$7 \cdot 3x \equiv 4 \pmod{10} \cdot 7$$

$$x \equiv 28 \pmod{10}$$

$$\equiv 8$$

Solving Linear Congruents

$$ax \equiv b \pmod{m}$$

Case 1

$$\text{GCD}(a, m) = 1$$

← unique solution
multiply by inverse of a

Case 2

$$\text{GCD}(a, m) = d > 1 \quad (i) \quad d \nmid b = \text{no solutions}$$

(ii) $d \mid b = \text{divide } d \text{ out, go back}$
to case 1

(8)

$$3x \equiv 13 \pmod{12} = \text{no solutions}$$

$$\text{GCD}(3, 12) = 3$$

$$3 \nmid 13 = \text{FALSE}$$

by case 2, no solutions

(9)

$$3x \equiv 9 \pmod{12}$$

$$x \equiv 3 \pmod{4}$$